Matrix Theory

06 Vector Spaces and Subspaces

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Beyond Elimination

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- Helps us with the basic questions of *existence* and *uniqueness*
- How to find every solution?

What makes Linear Algebra work?

- Fundamental operations on vectors?

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- Addition and scalar multiplication

Do vectors need to be columns of numbers?

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- Consider polynomials: $p(x) = 1 + 2x$ and $q(x) = -x^2$
- Add them: $p(x) + q(x)$
- Multiply by scalars: $3p(x)$

Do vectors need to be columns of numbers?

$$\checkmark \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad A$$

- Consider 2×2 matrices

$$\checkmark \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \quad B$$

$$D = c \cdot A + d B$$

Do vectors need to be columns of numbers?

- Consider 2×2 matrices
- Matrices can be added and multiplied by scalars

Do vectors need to be columns of numbers?

In linear algebra, a vector is not necessarily an arrow or a column of numbers. A vector is an object belonging to a set in which addition and scalar multiplication behave in the familiar linear way.

Vector Space

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

$\mathbb{R}$

Let V be a nonempty set, and let the scalars come from a field $\mathbb{F}$, (usually $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$)

We have two operations:
$+ : V \times V \to V$, called vector addition, and
$\cdot : \mathbb{F} \times V \to V$, called scalar multiplication.
We call V a vector space over $\mathbb{F}$ if, for every $u, v, w \in V$ and every $a, b \in \mathbb{F}$, the following properties hold:

Vector Space

1. Commutativity of addition $\quad u+v = v+w$
2. Associativity of addition $\quad u+(v+w) = (u+v)+w$
3. Existence of a zero vector $\quad u+0 = u$
4. Existence of additive inverses $\quad u+\underline{(-u)} = 0$
5. Distributivity over vector addition $\quad a(u+v) = au+av$
6. Distributivity over scalar addition $\quad (a+b)u = au+av$
7. Compatibility of scalar multiplication $\quad (ab)u = a(bu)$
8. Identity scalar $\quad 1.v = v$

Vector Space

There is an important point: $u + v \in V$ and $av \in V$.
That is, the set is closed under addition and scalar multiplication.

Vector Space

Addition behaves normally

$u + v = v + u$, $(u + v) + w = u + (v + w)$, $v + 0 = v$, $v + (-v) = 0$

Scalar Multiplication behaves normally

$1v = v$, $a(bv) = (ab)v$

Addition and Scalar Multiplication interact normally

$a(u + v) = au + av$, $(a + b)v = av + bv$

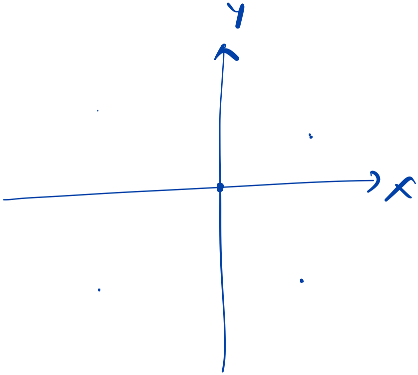
Vector Space

Informally

A vector space is a collection of objects in which linear combinations make sense and stay inside the collection.
If, $v_1, v_2, \ldots v_k \in V$, and
$c_1, c_2, \ldots, c_k \in \mathbb{F}$, then $c_1 v_1 + c_2 v_2 + \ldots + c_k v_k \in V$.

Example Vector Spaces

- $\mathbb{R}^2, \mathbb{R}^3, \underline{\mathbb{R}^n}$



Example Vector Spaces

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- $M_{m\times n}(\mathbb{R})$

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Example Vector Spaces

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- $C[a, b] = \{\text{Continuous functions in } [a, b]\}$ ✓

$f(x) = \sin x$

$g(x) = e^{-x}$

Example Vector Spaces

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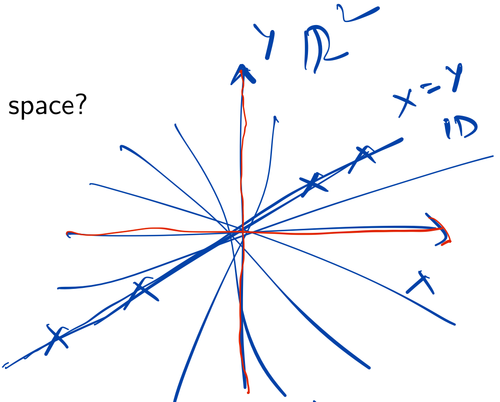
Vectors in Vector Space

These objects look completely different - columns, matrices, polynomials, functions - but from the viewpoint of linear algebra, they are all the same kind of object.

Subspace

$\left\{\begin{pmatrix}0\\0\end{pmatrix}\right\}$

- Is every subset of $\mathbb{R}^n$ a vector space?



(x, y)

(x, y)

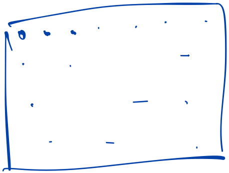
$x_1 = y_1$

$a(x_1 \ y_1) + b(x_2 \ y_2) = (ax_1 + bx_2 \quad ay_1 + by_2)$

$x_2 = y_2$

Subspace

- Is every subset of $\mathbb{R}^n$ a vector space?
- A subspace is a subset that is ‘closed’ under addition and scalar multiplication



Image

12×10^{1} pixels

High-dim data may lie on a low-dim subspace!

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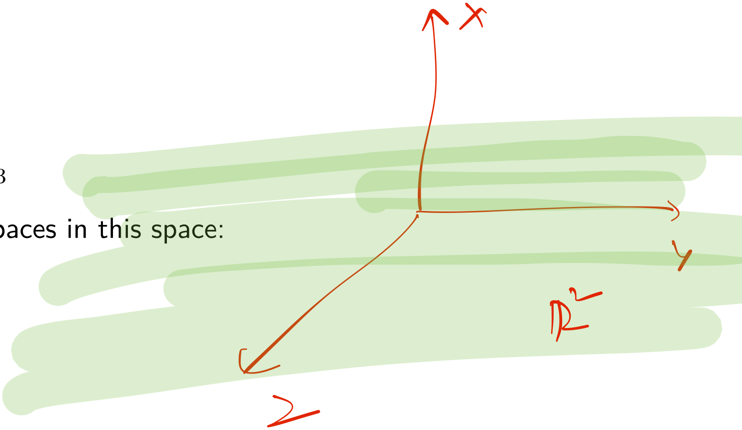
Subspace

- Is every subset of $\mathbb{R}^n$ a vector space?
- A subspace is a subset that is 'closed' under addition and scalar multiplication
- Zero vector belongs to every subspace
- Smallest subspace of a space?
- Largest subspace of a space? $\mathbb{R}^n$

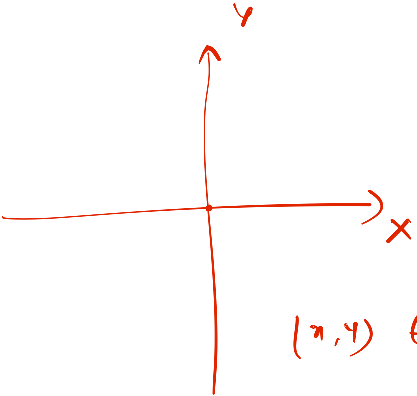
Subspace

- Consider $\mathbb{R}^3$
- Some subspaces in this space:



Subspace

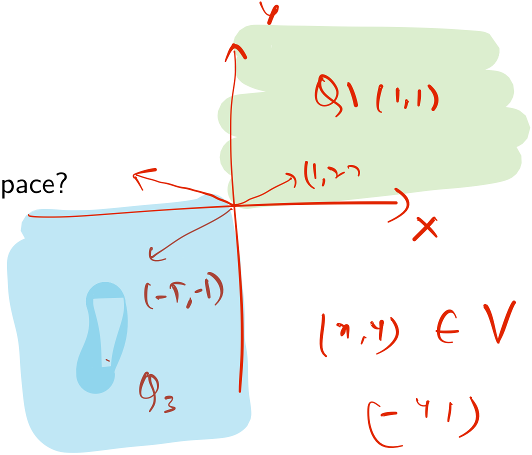
- Consider $\mathbb{R}^2$



$(x, y) \in \mathbb{R}^2$

Subspace

- Consider $\mathbb{R}^2$
- Is the first quadrant a subspace?



$(x, y) \in V$

$(-4, 1)$

what about $Q_1 \cup Q_3$?

Next

- Column space and Null space of a matrix